

Multivariate statistics

Lecture 09:

Introduction to non-parametric statistics and hypothesis tests

Literature and Problems: See course schedule

Introduction to the lecture

Many methods in classical statistics and also more than half of this course is based on an assumption that the collected data sample come from a (multivariate) normally distributed random variable. This assumption is often justified with reference to the Central Limit Theorem and also works well for many real-life cases.

However, for some data sets it is not possible to assume even an approximate normal distribution, which means that the classical methods break down.

In this lecture will be given a short introduction to non-parametric statistics and in particular non-parametric hypothesis tests. The term “non-parametric” basically means that no assumption of distribution is made.

Due mostly to limited time available, but also for simplicity, focus will be on univariate non-parametric hypothesis tests. However, the basic principles can to some degree be generalized to a multivariate setting – not surprisingly with increased complexity. Thus, along with the material used in the lecture, a few papers on multivariate generalizations will be available.

It should be noted that if the data set actually can be reasonably approximated to a normal distribution, the classical methods and hypothesis tests will have higher power than the non-parametric tests of this lecture.

Agenda (preliminary):

Location tests

Location tests is about testing which value “most” of the data sample is centered around. In classical statistics this means testing for mean values, where in non-parametric tests the more robust measure the median is used.

One sample or paired data

- Sign test (SLIDE + MATLAB demo)
- Wilcoxon signed rank test (SLIDE + MATLAB demo)

Two samples

- Mann-Whitney U-test (SLIDE + MATLAB demo)

Multiple sample

- Kruskal-Wallis test (ANOVA1) (SLIDE + MATLAB demo)
- Friedman test (ANOVA”2”) (SLIDE + MATLAB demo)

Contingency table test for categorical data

Contingency tables consist of counts of how many observations belong to combinations of two specific factors (categories) and can be used to test for independence of the factors.

- Contingency table test (SLIDE + MATLAB demo)

Distribution tests

Distribution tests is used to test whether a data set might plausible come from a random variable with a specified (assumed) distribution with either known or unknown parameters. or alternatively to test whether two data sets might originate from identical distributions.

- One sample Kolmogorov-Smirnov test (SLIDE + MATLAB demo)
- Two-sample Kolmogorov-Smirnov test (SLIDE + MATLAB demo)
- Goodness of fit test (SLIDE + MATLAB demo)